TASK -1

FOUNDATIONS OF CYBERSECURITY

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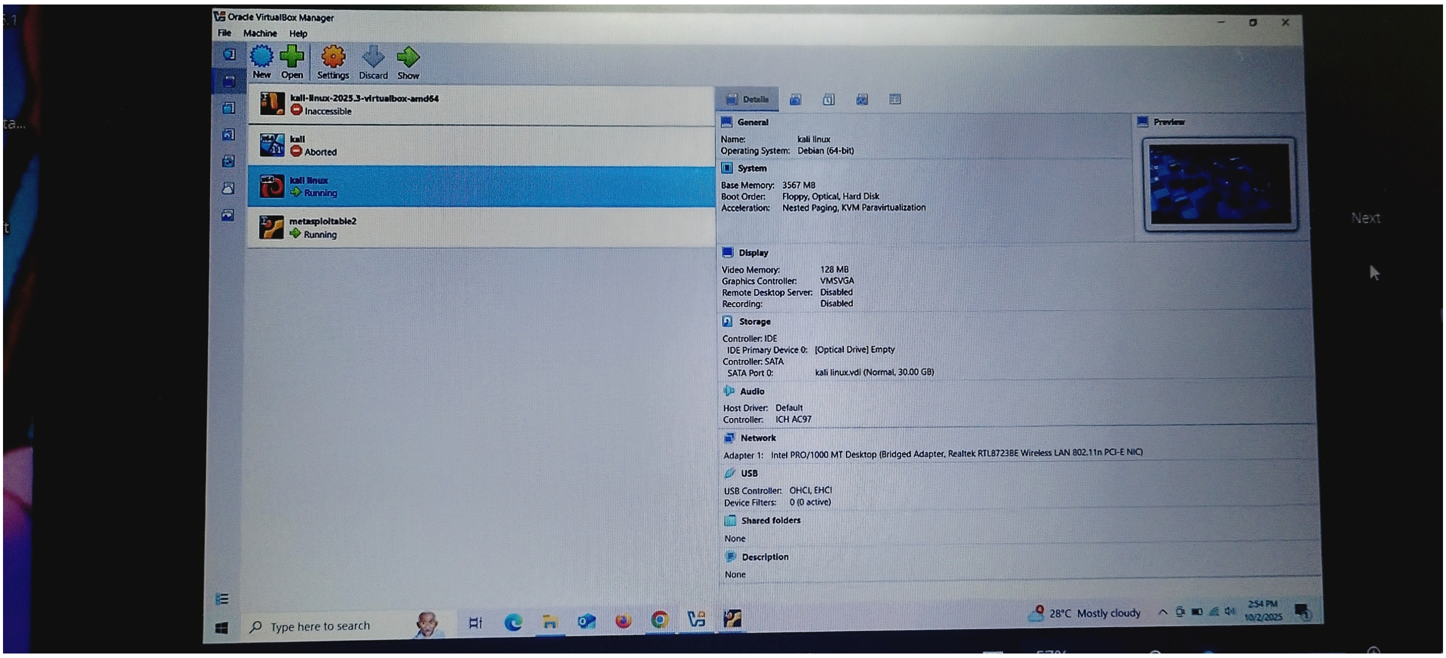
- **OBJECTIVE**

To build a cybersecurity lab and learn basic Linux, networking, cryptography, and tool usage.

- **LAB SETUP**

I installed Kali Linux (attacker) and Metasploitable 2 (target) in VirtualBox and configured Host-Only network.



- **LINUX FUNDAMENTALS**

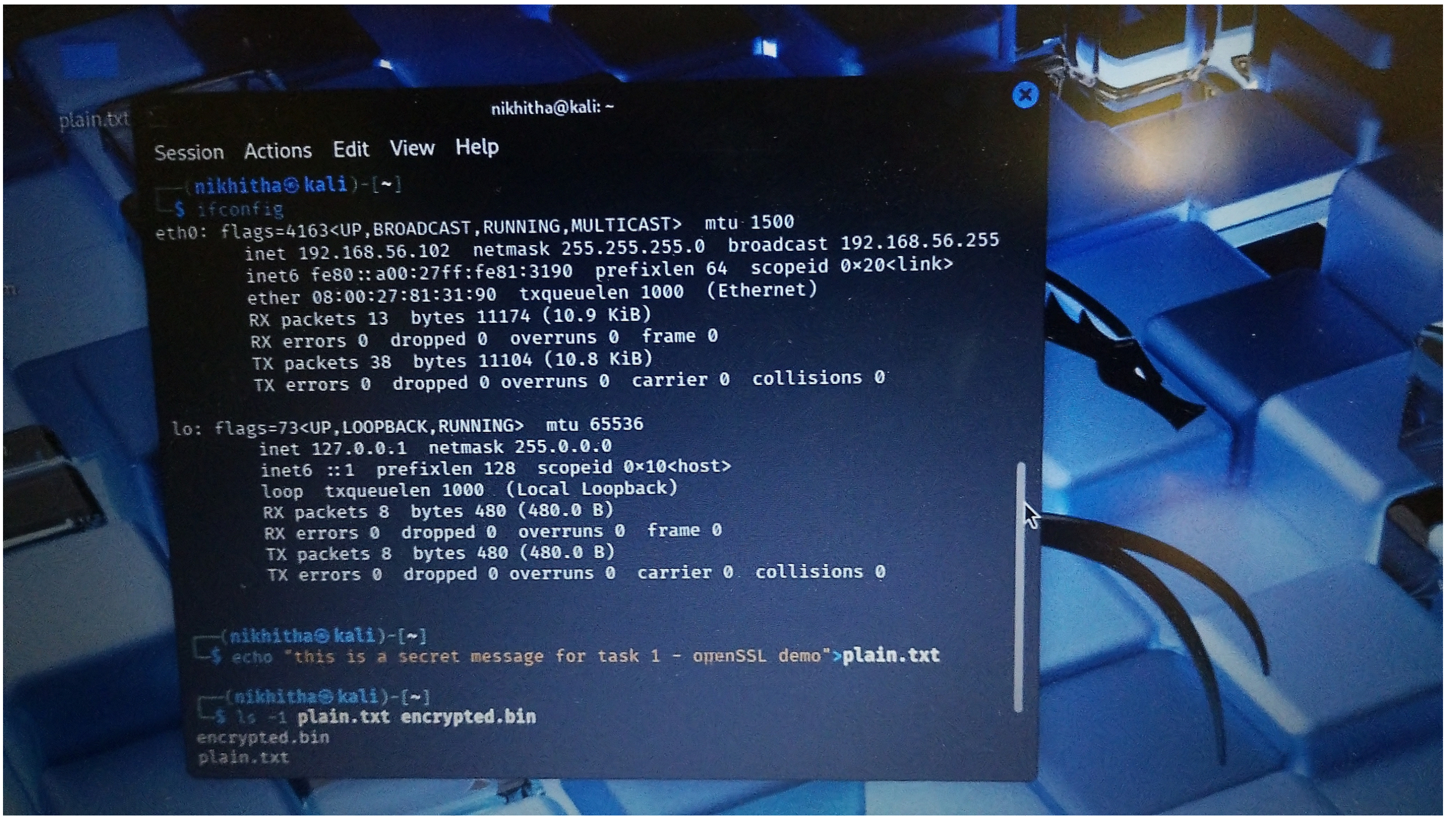
-PWD: Shows the current working directory

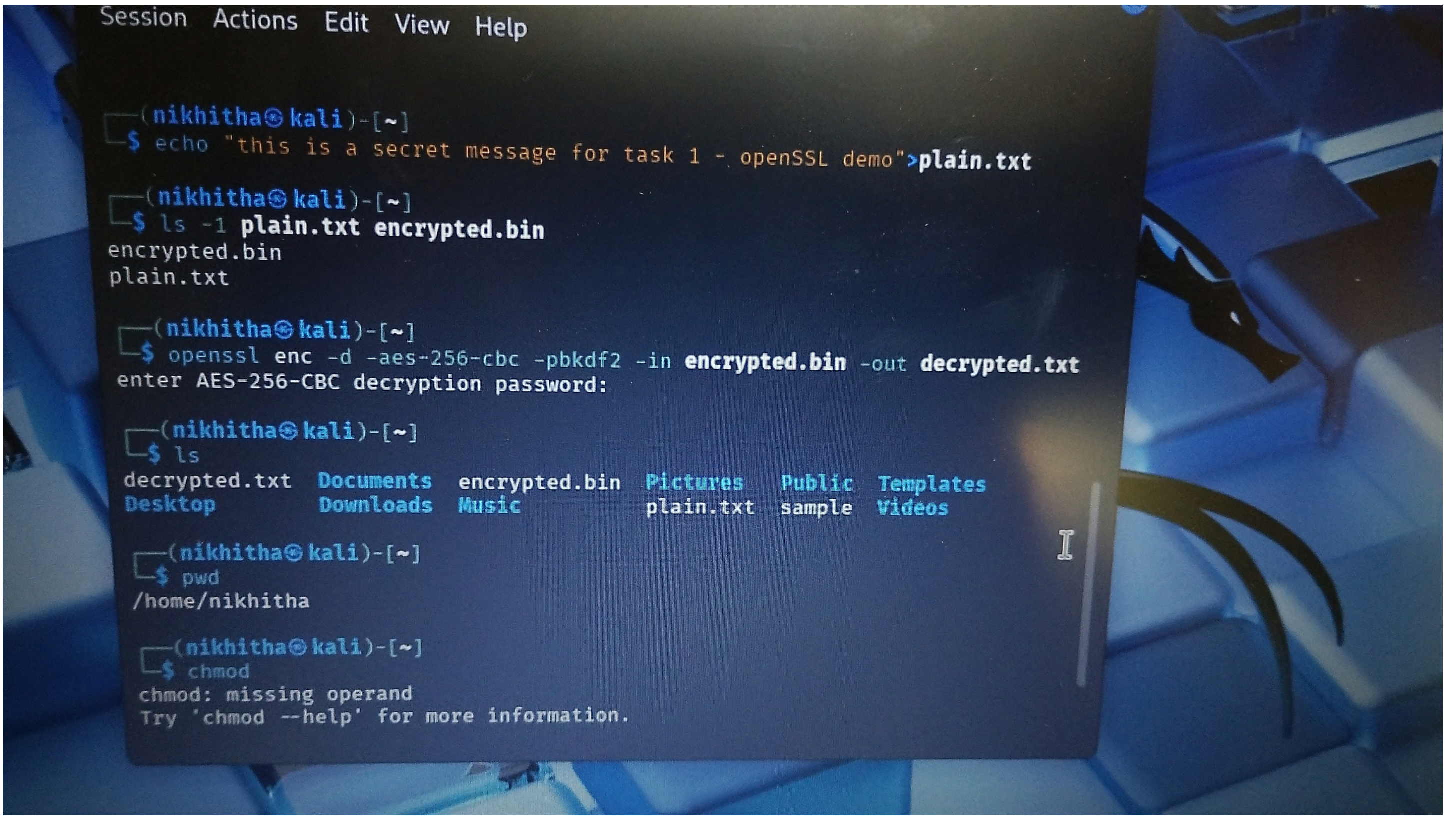
-LS: lists files and directories

-IFCONFIG: shows the ip addresses

-TOUCH: creates a new file

-APT UPDATE: updates the package list



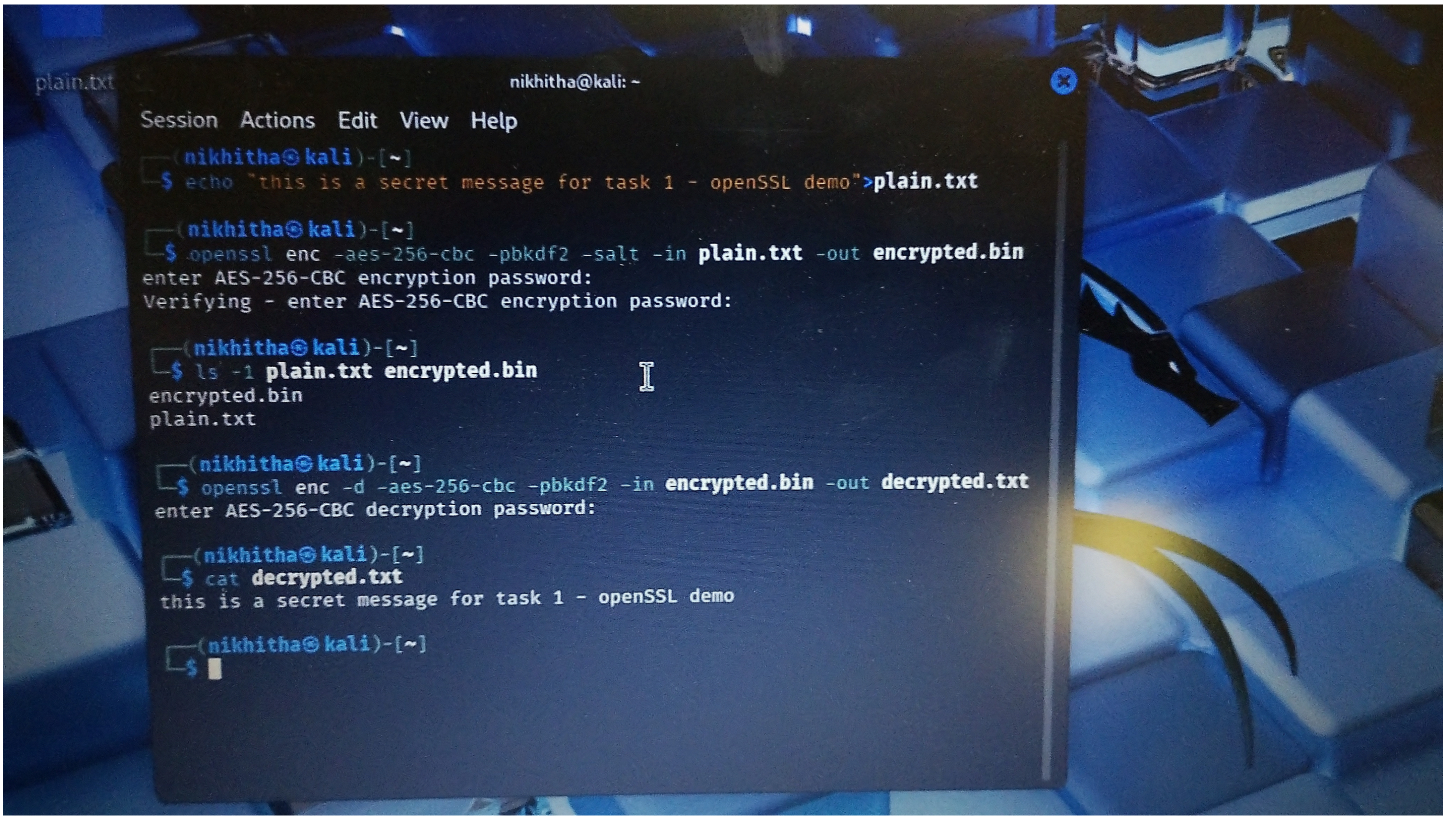


- ***OPENSSL DEMO(CRYPTOGRAPHY)***

ENCRYPTED COMMAND: echo "This is a secret message for Task 1 - OpenSSL demo" > plain.txt

DECRYPTED COMMAND: openssl enc -d -aes-256-cbc -pbkdf2 -in encrypted.bin -out decrypted.txt

OUTPUT:

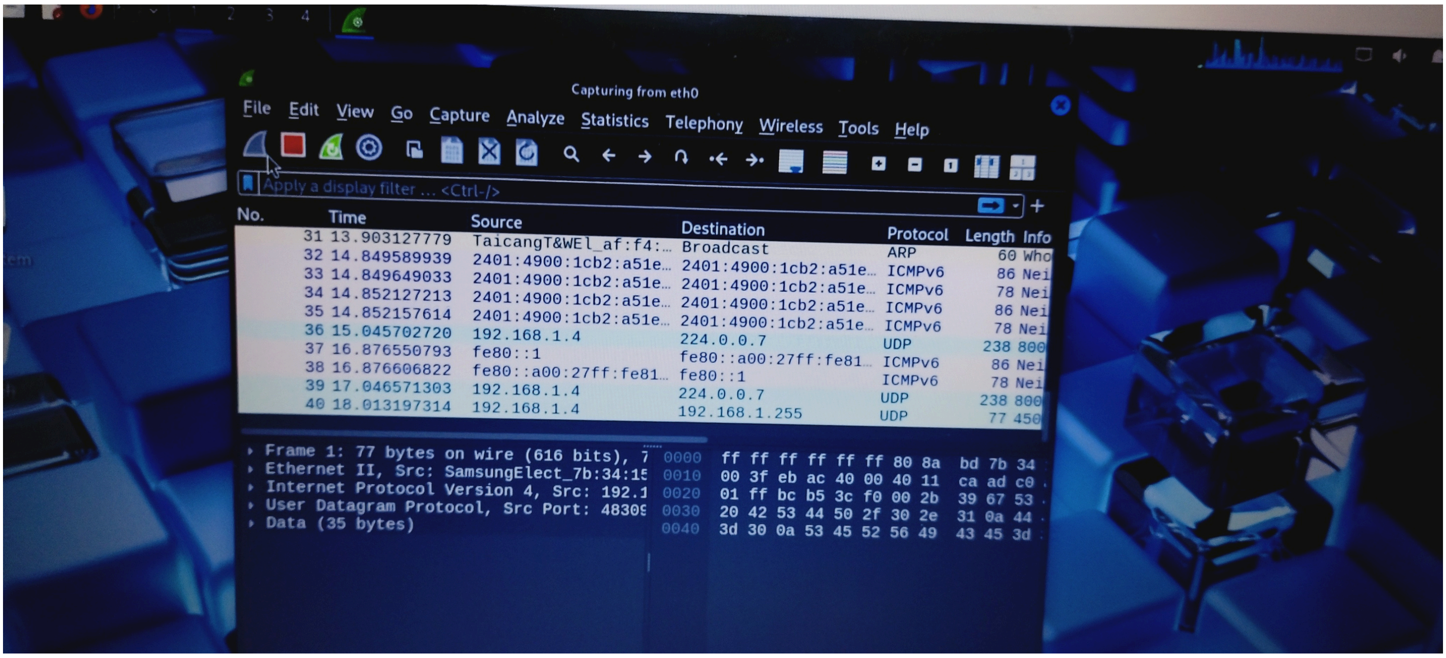


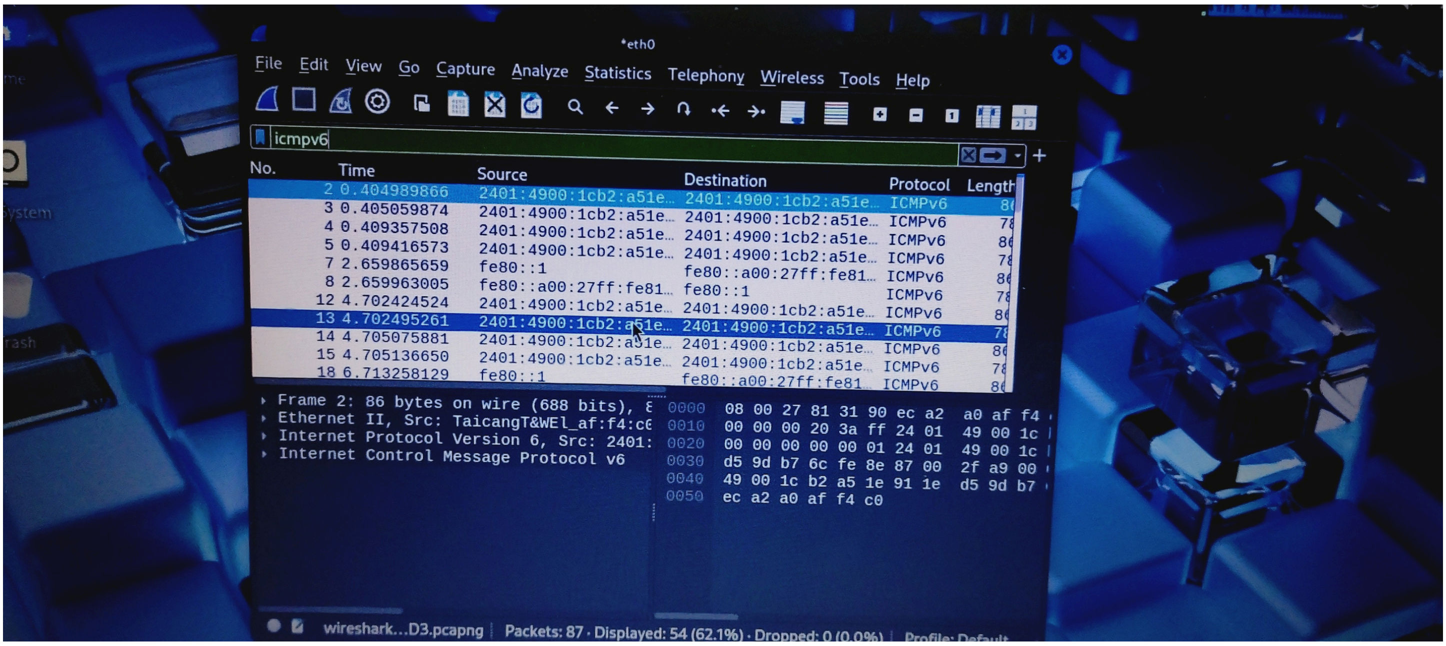
- ***TOOL FAMILIARIZATION***

Wireshark — Packet capture & analysis

What it does: Captures network traffic and displays packet-level details (source/destination IP, protocols, payloads). Useful for spotting unencrypted credentials and protocol problems.

How I used it: Captured traffic while pinging the target VM and filtered HTTP traffic.



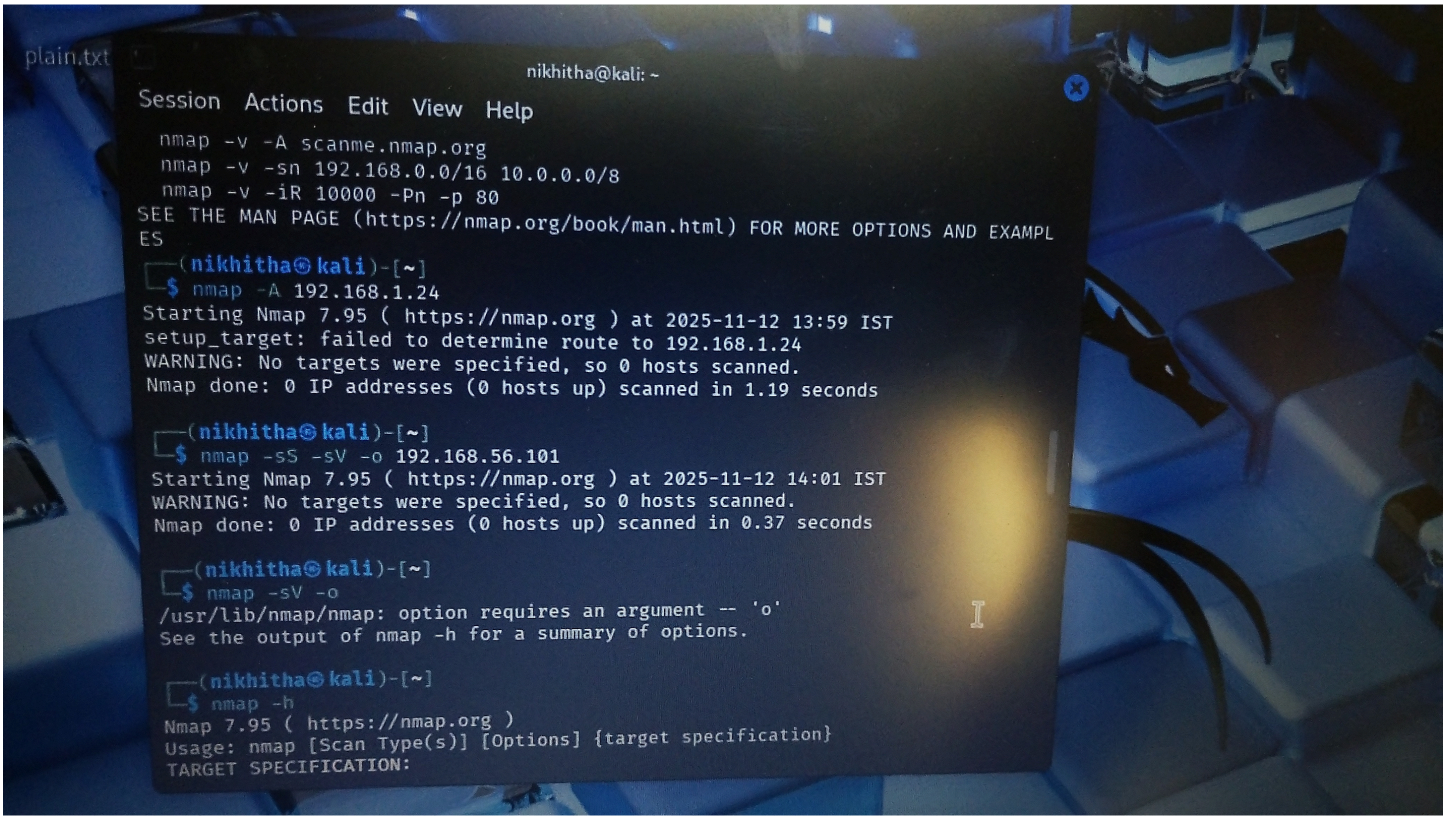


Nmap — Network & port scanner

What it does: Discovers hosts, open ports, services, and OS details on a network.

How I used it: Ran service/version and OS detection on the target VM.

Sample command: nmap -sS -sV -O 192.168.56.101

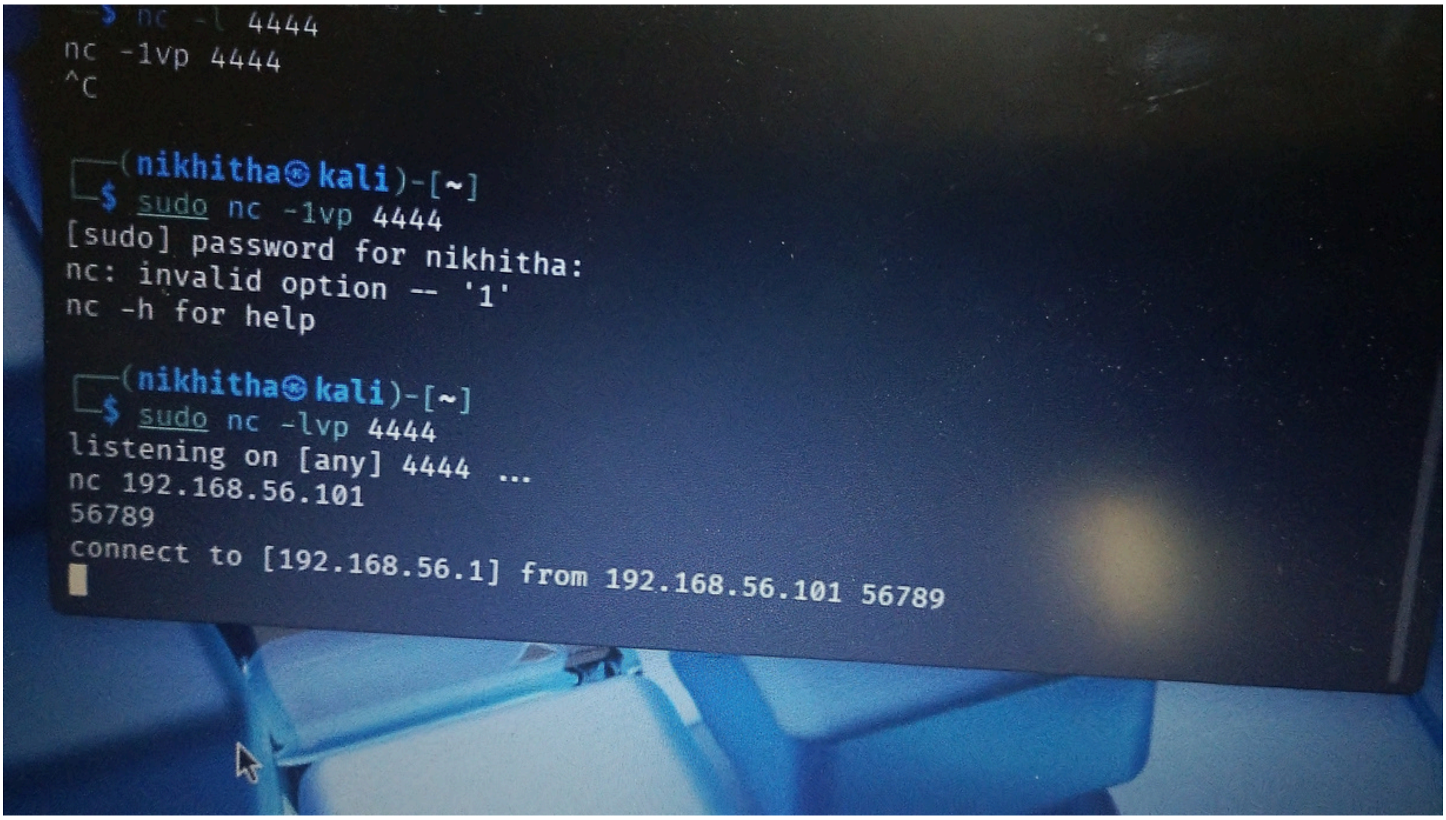


Netcat (nc) — Network debugging / simple server & client

What it does: Read/write data across network connections; useful for quick listeners/reverse shells in labs.

How I used it: Listened on a test port and sent a small message from the target.

Sample command: nc -lvp 4444 (listener)



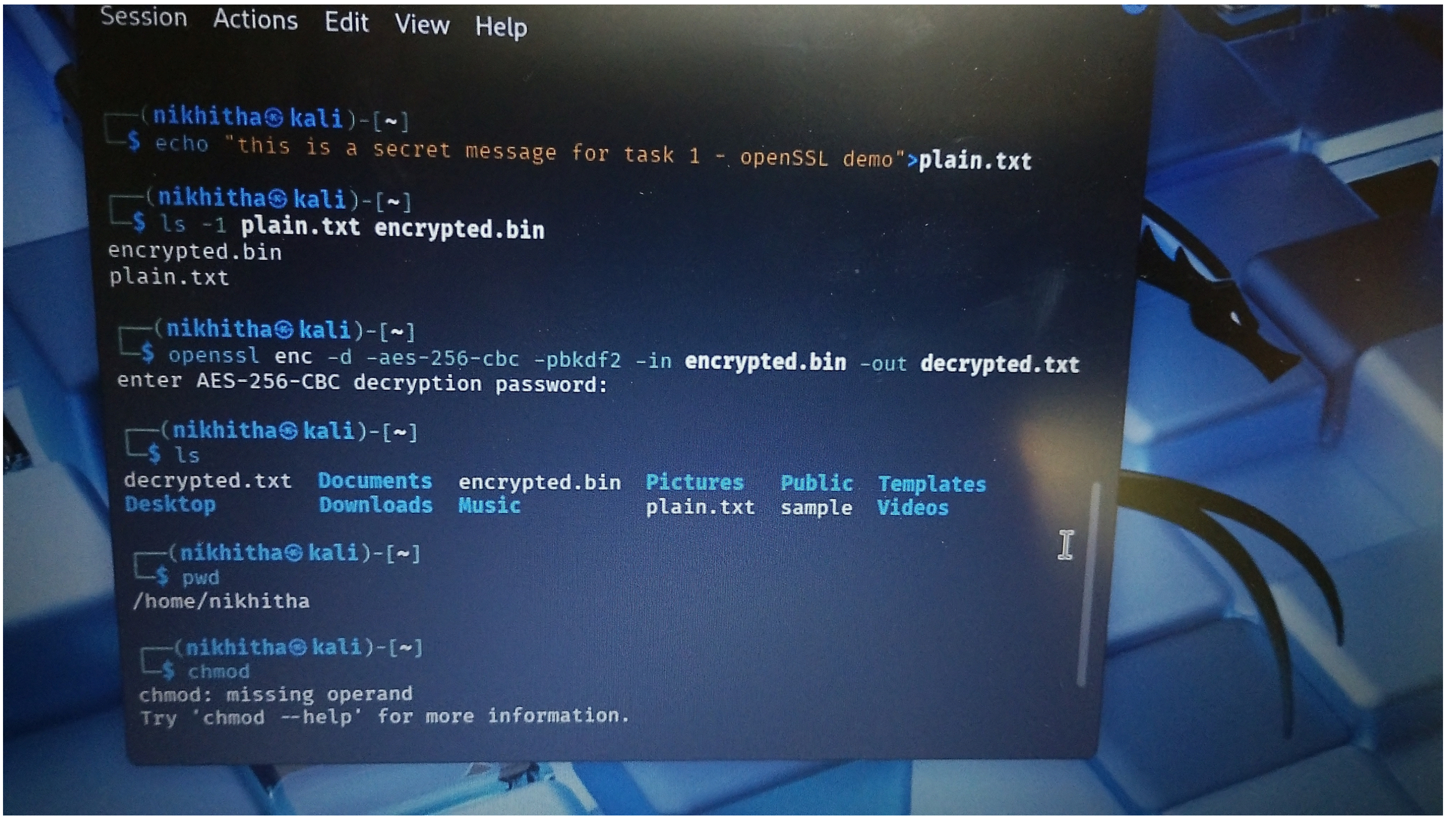
OpenSSL — Cryptography toolkit

What it does: Encrypt, decrypt, sign, and verify files; generate keys and certificates.

How I used it: Performed AES-256 symmetric encryption and RSA public/private encryption.

Sample command:

openssl enc -aes-256-cbc -pbkdf2 -salt -in plain.txt -out encrypted.bin



- **WHAT I LEARNED**

During Task 1 I successfully built a local cybersecurity lab and practiced essential Linux and networking commands. I learned to install and operate Kali Linux and a vulnerable target VM, configure a host-only network, and verify connectivity using ping. I gained hands-on familiarity with important tools: Nmap for discovery and reconnaissance, Wireshark for packet capture and analysis,and OpenSSL for cryptographic operations (AES symmetric encryption and RSA key operations). Practically performing encryption and decryption helped me understand key management and the differences between symmetric and asymmetric cryptography. Overall, this task strengthened my foundational skills required for secure system administration and penetration testing.